

Hamiltonian Neural Networks for Robust Out-of-Time Credit Scoring: Empirical Validation and Temporal Stability Analysis

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Abstract—This paper presents a novel credit scoring approach using neural networks inspired by Hamiltonian mechanics to address class imbalance and out-of-time prediction challenges. We develop a specific optimizer and loss function that incorporates physical system principles to better capture credit risk dynamics. Testing on the Freddie Mac Single-Family Loan-Level Dataset demonstrates our model achieves superior discriminative power (AUC scores of 0.8027 vs. 0.6072) in out-of-time scenarios compared to conventional methods, despite lower traditional accuracy metrics. Statistical significance testing confirms these improvements are not due to random variation. Our empirical validation through ablation studies reveals that both Hamiltonian normalization and momentum updates contribute significantly to performance, with their combined effect representing an 11.51% improvement over standard neural networks. The approach demonstrates remarkable temporal stability, maintaining consistent performance between in-sample and future test sets, with minimal degradation across increasing time horizons (12, 36, and 60 months) compared to standard techniques which show steeper performance decline. This interdisciplinary method bridges physical systems theory and financial risk management, offering practical advantages for scenarios where accurate risk ranking and long-term model stability outweigh simple classification accuracy.

Index Terms—Credit Scoring, Out-of-Time Prediction, Class Imbalance, Hamiltonian Neural Networks

I. INTRODUCTION

Credit scoring models have significant challenges with maintaining predicted accuracy over time, primarily because of fluctuating economic conditions and changing data distributions [2]. Conventional credit-scoring models have been assessed in static environments, using randomly partitioned datasets for training and validation [15]. This approach overlooks the shifting nature of financial data and the possibility of concept drift in credit risk factors [34]. The importance of out-of-time (OOT) testing in financial models is critical. Out-of-Time (OOT) testing uses a forward-looking methodology in which adjacent, non-overlapping sample intervals assess model accuracy [32]. This approach guarantees that models are evaluated on data representative of future conditions, determinant when measuring a model's prediction accuracy in practical situations [20].

OOT hasn't always been effectively used in credit scoring research, mostly due to the lack of long-term datasets [28]. The Freddie Mac Single-Family Loan-Level Dataset (SFLLD) fills this void by offering comprehensive historical data, facilitating thorough out-of-time testing [28]. Using data from multiple periods for training and testing enables researchers to more

accurately evaluate a model's robustness to fluctuations in economic conditions and consumer behaviour [21]. Moreover, credit scoring models must address class imbalance, as instances of default are generally few relative to non-default situations [19]. This disparity may result in biased models that underperform on minority classes, requiring particular methodologies for model building and assessment [23].

Addressing these difficulties requires novel approaches that can effectively represent the temporal dynamics of credit risk while preserving robustness against class imbalance and concept drift. Our proposed Hamiltonian-inspired neural network attempts to address these challenges by using principles from physics to capture the complex, time-dependent nature of credit risk.

A. Key Contributions

This paper makes several significant contributions to the field of credit scoring and financial risk modeling:

- A novel physics-inspired neural network optimization framework specifically designed for credit risk assessment that leverages Hamiltonian mechanics principles to maintain temporal stability
- Empirical evidence demonstrating superior discriminative power (AUC) in out-of-time scenarios, even when conventional accuracy metrics favor traditional approaches
- A new perspective on the connections between physical system dynamics and financial risk evolution, providing theoretical justification for the observed stability improvements
- An adaptable framework that prioritizes risk ranking capabilities over simple binary classification, aligning with practical needs in financial decision-making

II. HAMILTONIAN MECHANICS IN NEURAL NETWORKS OPTIMIZATION

Optimization of Neural Networks (NN) is essentially a high-dimensional non-convex optimization problem [16]. Existing optimization techniques for neural networks, including Stochastic Gradient Descent (SGD) and its derivatives, seek to minimize a loss function through iterative updates of model parameters [30]. Yet, these techniques frequently face constraints such as slow convergence, trapping in local minima, and difficulties in avoiding saddle points [10].

Recent developments in neural network optimization focused on the geometry of the loss landscape [24] and the

dynamics of the optimization process [39]. The research findings have shown that the optimization trajectory of neural networks can be perceived as a dynamical system, thereby facilitating the application of concepts from physics and dynamical systems theory to neural network optimization [37]. A particularly promising approach stems from Hamiltonian mechanics, a reformulation of classical mechanics that offers a solid foundation for the analysis of dynamical systems [5]. The Hamiltonian formalism presents the notion of phase space, a key structure in classical mechanics and current optimization theory [27].

The Hamiltonian formalism provides an effective mathematical framework for developing conservative mechanical system theory, and is a geometric language for multiple fields of physics [12]. The Hamiltonian formalism introduces phase space, a $2n$ -dimensional space with n degrees of freedom. Each point in phase space represents an individual state, described by its position and momentum coordinates (q, p) . The equations of motion in a Hamiltonian system $H : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ can be defined as

$$\frac{dq}{dt} = \frac{\partial H}{\partial p} \quad (1)$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q} \quad (2)$$

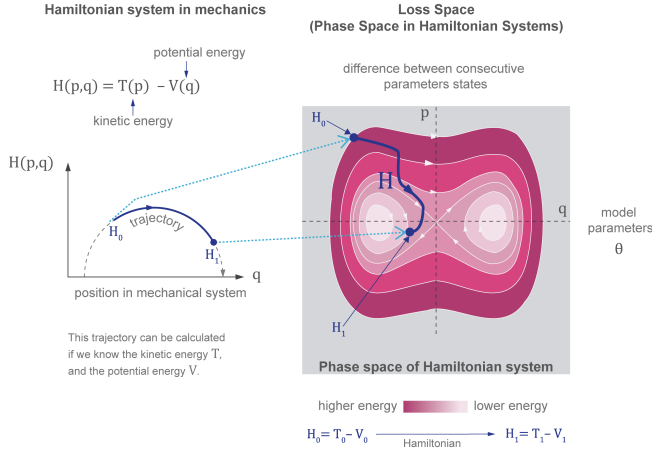


Fig. 1. Equivalence between Hamiltonian systems in physical systems, and its application in NN optimization spaces. Position q is equivalent to the NN parameters θ , and momentum vector p is equivalent to the difference between consecutive parameters states.

In many real-world situations, these equations can't be solved analytically. Symplectic integrators are calculation methods specifically formulated for solving Hamiltonian systems while preserving their fundamental geometric features, particularly the symplectic structure. Symplectic structures are fundamental geometric objects in differential geometry and classical mechanics, and support Hamilton's equations of motion by explaining the connection between position and momentum in physical systems [27]. In simple terms, symplectic structures are specific rules that define how things

move in physics, similar to an equation for motion. We can define a Hamiltonian in an optimization space system as:

$$H(q, p) = T(p) - V(q) \quad (3)$$

Where q represents the current state of the model parameters (θ), analogous to position in mechanical systems, and p is the difference between consecutive parameter states, $p_t = \theta_{t+1} - \theta_t$, and is analogous to the momentum vector in mechanical systems. In equation (3), $T(p)$ is the 'kinetic energy' term representing the cost of changing the model parameters, and $V(q)$ is the 'potential energy' term representing the loss function of the current model state [5], [11], [37].

Symplectic integrators are numerical methods specifically formulated to solve Hamiltonian systems while preserving their fundamental geometric features, particularly the symplectic structure [17]. These integrators are needed for long-term simulations of Hamiltonian systems as they preserve important physical quantities, including energy and angular momentum, with minimal error over long intervals [26]. In Hamiltonian systems, a symplectic integrator approximates the flow ϕ_t of Hamilton's system defined by equations (1-2). Symplectic integrators are characterized by their preservation of the symplectic two-form $\omega = \sum dq_i \wedge dp_i$, which corresponds to the conservation of phase space volume in Hamiltonian systems [31]. This property guarantees the preservation of the qualitative features of the original continuous dynamical system, even with long step sizes [31]. Some examples of integrators are the Störmer-Verlet method, commonly referred to as the 'leapfrog method' in molecular dynamics simulations, is a frequently used symplectic integrator [36]. Fourth-order symplectic integrators, like the Forest-Ruth algorithm, offer enhanced long-term stability along with better energy conservation properties [14]. One of the simplest forms of symplectic integration is the first-order symplectic integrator called the Symplectic Euler method [22]. For a Hamiltonian system with position q and momentum p , the Symplectic Euler method can be represented as:

$$p_{t+1} = p_t - \Delta t \times \frac{\partial H}{\partial q}(q_t, p_{t+1}) \quad (4)$$

$$q_{t+1} = q_t - \Delta t \times \frac{\partial H}{\partial p}(q_t, p_{t+1}) \quad (5)$$

where H is the Hamiltonian and Δt is the time step, analogous to the learning rate in a non-convex optimization problem. In this paper, we proposed a hybrid approach that combines the basic structure of the Symplectic Euler method, momentum-based updates [33], and a Hamiltonian-inspired normalization factor. This combination aims to use the advantages of each component: the preservation of the symplectic structure inherent in the Euler method, the better convergence features of momentum-based methods, and the energy conservation principles derived from Hamiltonian mechanics [25]. Our hybrid approach uses a momentum update similar to Adam or RMSprop optimizers. We can represent it as:

$$m_t = \beta \times m_{(t-1)} + (1 - \beta) \times g_t \quad (6)$$

Require: Learning rate η , momentum coefficient β , epsilon ϵ

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1: for each iteration do
2:   for each parameter  $\theta$  in model parameters do
3:      $g \leftarrow \nabla L(\theta)$  {Compute gradient}
4:      $v \leftarrow \text{state}[\theta]['\text{momentum}']$  {momentum}
5:     {Update momentum}
6:      $v \leftarrow \beta \times v + (1 - \beta) \times g$ 
7:     {Compute Hamiltonian}
8:      $K \leftarrow 0.5 \times \|v\|^2$  {Kinetic energy}
9:      $V \leftarrow 0.5 \times \|\theta\|^2$  {Potential energy}
10:     $H \leftarrow K + V$  {Hamiltonian}
11:    {Update parameter}
12:     $\theta \leftarrow \theta - \eta \times v / (\sqrt{H} + \epsilon)$ 
13:    {Store updated momentum}
14:     $\text{state}[\theta]['\text{momentum}'] \leftarrow v$ 
15:  end for
16: end for

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Fig. 2. Symplectic Optimizer

Where m is the momentum, β is a decay factor, and g_t is the gradient. Our parameter update includes a Hamiltonian normalization:

$$\theta_t = \theta_{t-1} - \frac{\eta \times m_t}{\sqrt{H_t} + \epsilon} \quad (7)$$

Then we compute the Hamiltonian as the sum of kinetic and potential energy $H(q, p) = T(p) + V(q)$. Our approach preserves the idea of symplectic integration by separating momentum (6), and position (7) updates. These approaches have been shown to accelerate convergence and help overcome local minima in neural network training [33]. The normalization by $\sqrt{H_t} + \epsilon$ in (7) is special, and not part of standard symplectic integrators (ϵ is a small constant for stability). This is derived from the energy conservation principles in Hamiltonian mechanics, and aids in adaptively adjusting the step size based on the "energy" of the system, potentially leading to more stable and efficient optimization [25]. The momentum decay factor β is borrowed from modern optimization techniques.

The advantages of our proposal are the following: better energy conservation, improved long-term stability, and potential for larger learning rates. Conventional loss functions in neural networks, such as cross-entropy, primarily aim to minimize prediction errors [16]. In the complex optimization landscapes characteristic of deep learning, these functions may fail to accurately represent the geometrical properties of the parameter space or enable stable learning dynamics [24]. Our Hamiltonian-inspired loss function covers these limitations by integrating principles from classical mechanics.

We define the loss function as:

$$H_{\text{loss}}(\theta) = L_{\text{base}}(\theta) + \lambda \times R(\theta) \quad (8)$$

where L_{base} represents the standard loss (e.g., cross-entropy), $R(\theta)$ denotes a regularization term analogous to potential energy in Hamiltonian systems, and λ is a hyperparameter that controls the strength of regularization [5].

The regularization term $R(\theta) = (1/2)\|\theta\|^2$ penalizes large parameter values, leading to more accurate solutions. This formulation is derived from the potential energy component in Hamiltonian mechanics, $V(q) = (1/2)\|p\|^2$, where p denotes momentum [27].

Through the incorporation of this physics-inspired regularization, our loss function seeks to enhance generalization by promoting more uniform parameter distributions [29]. At the same time, we can improve optimization stability by applying the energy conservation principles inherent in Hamiltonian systems [37]. With this approach, we provide a natural framework for balancing model complexity and data fit, analogous to the connection between kinetic and potential energy in physical systems.

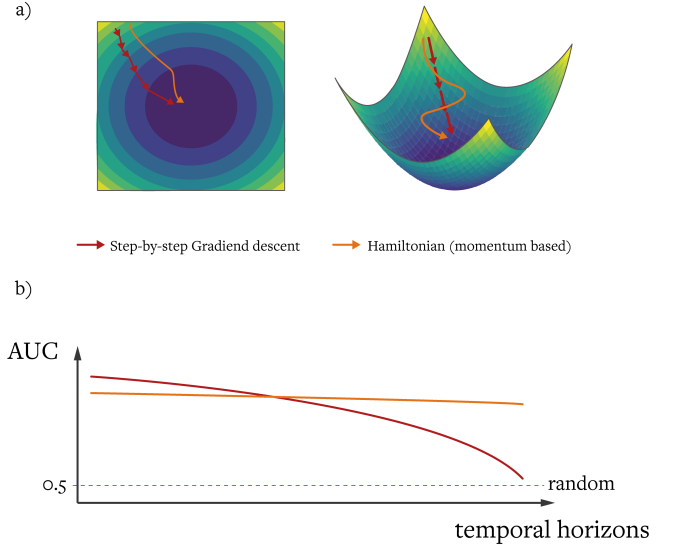


Fig. 3. Optimization dynamics and temporal stability properties of Hamiltonian neural networks. (a) Comparison of optimization trajectories in loss landscape between standard gradient descent (red, left) which converges directly to local minima, and Hamiltonian momentum-based approach (orange, right) which maintains orbital trajectories that explore a broader region of parameter space. (b) The consequence of these different optimization behaviors on model performance stability over time, showing how Hamiltonian approaches maintain discriminative power (AUC) with minimal degradation across increasing time horizons compared to standard methods which exhibit steeper performance decline. This stability results from the energy conservation principles inherent in Hamiltonian systems that prevent overfitting to specific temporal patterns.

A. Linking Hamiltonian Dynamics to Financial Risk Evolution

Financial systems, like physical systems, have dynamics that can be characterized by conservation laws and state transitions. Credit default events can be conceptualized as state transitions in a complex dynamical system with temporal dependencies. Traditional machine learning approaches often fail to capture these underlying dynamics, particularly when economic conditions fluctuate.

The mathematical structure of financial time series has three basic analogies with Hamiltonian systems. First, though financial markets lack true energy conservation in the thermodynamic sense, they exhibit conservation-like regularities in risk

propagation—information neither created nor destroyed, but transformed through complex market interactions, manifesting as invariant properties across temporal scales. Second, economic systems undergo remarkable critical transitions, where microscopic fluctuations suddenly amplify into macroscopic restructuring during financial crises, analogous to how phase space bifurcations in physical systems lead to qualitative state changes under parametric variation. Third, financial systems encode memory in their trajectory—a kind of path-dependent momentum where historical states constrain future possibilities through institutional structures, psychological anchoring, and regulatory frameworks.

Our framework explicitly models the optimization space as a physical manifold with dual energetic components: kinetic energy and potential energy. The kinetic term—representing the momentum of parameter updates—interacts with the potential term—embodying the loss function’s topography—to create a natural equilibrium between model complexity and data fit.

III. DATA AND METHODOLOGY

We use the FMDData dataset introduced by Mushava & Murray [28]. This dataset originates from the Freddie Mac Single-Family Loan-Level Dataset (SFLLD) and includes mortgage loans granted in the USA, along with their subsequent repayment patterns. It contains a list of fully amortized fixed-rate mortgage loans. The data covers the period from 1999 to a more recent date, based on the access date, and includes both loan origination data and monthly performance metrics. A unique loan identifier is used to gather loan origination and repayment information. Default is represented by the failure to make three or more payments within defined performance intervals of 12, 24, 36, 48, and 60 months.

In our experiment, we use the out-of-time (OOT) dataset from FMDData, intended to evaluate model performance on future data through non-overlapping temporal intervals for training and testing. The OOT test set originates from a period following the training set data. It attempts to recreate real-world scenarios in which models forecast future, unobserved data. This dataset includes performance windows of 12, 36, and 60 months. This data offers a practical credit scoring scenario, enhancing the relevance of our research to real financial risk analysis. The OOT dataset enables the assessment of our model’s performance over time, which is essential in credit scoring due to fluctuating economic conditions. Using out-of-time data evaluates our model’s capacity to generalize to future, unobserved data – an essential criterion in practical applications. The data from Freddie Mac is highly valued in the financial sector, increasing the credibility of our research.

Our experimental setup is designed to carefully evaluate the effectiveness of our Hamiltonian-inspired neural network methodology in credit scoring, with particular emphasis on how well it holds up in out-of-time (OOT) contexts. The configuration includes data partitioning, management of class imbalance, hyperparameter tuning, and a cross-validation approach. We split the FM Data into three sets: a training set

for model training that includes loans from the earliest time frame, a validation set for hyperparameter optimization and early termination of training, and a test set with loans from a subsequent time frame to ensure no temporal overlap. The OOT test set is essential for evaluating the model’s capacity to generalize to future, unseen data, simulating real-world credit scoring situations [21].

Credit default datasets often exhibit considerable class imbalance [23]. We implement the Synthetic Minority Over-sampling Technique (SMOTE) [7] on the training set to resolve this limitation. SMOTE generates synthetic instances of the minority class (defaults) within the feature space. It’s very important that we only implement SMOTE on the training data, thereby ensuring that the validation and out-of-time test sets retain their original distributions, thus preserving the integrity of our evaluation [9].

We use a grid search methodology for hyperparameter optimization [3]. The principal hyperparameters adjusted include: Learning rate (η), Momentum coefficient (β), Regularization intensity (λ), and network architecture (quantity and dimensions of hidden layers). Optimization involves using the validation set to prevent overfitting to the training data. We use a time-based cross-validation strategy to guarantee accurate performance estimation [1]. This methodology identifies the temporal patterns of the data. The training data is partitioned into K folds while preserving temporal sequence. For each fold k , we train on folds 1 through k and validate on fold $k + 1$. The procedure is executed $K - 1$ times, with the final fold designated for conclusive validation. This strategy offers a more accurate evaluation of the model’s performance over time, essential for credit scoring applications.

We perform a comparative analysis to benchmark the performance of our Hamiltonian-inspired neural network against XGBoost, an effective gradient boosting algorithm commonly used in credit scoring applications [8]. XGBoost has shown remarkable efficiency in many machine learning applications, especially in structured data problems such as credit risk evaluation [6]. We use XGBoost applying the same data partitioning, class imbalance management, and cross-validation methodology as our Hamiltonian-inspired model. This guarantees a fair comparison between the two approaches [38]. The XGBoost model is optimized using its specific hyperparameters, which include: number of estimators, maximum tree depth, learning rate, subsample ratio, and column sample ratio. Hyperparameter optimization for XGBoost is achieved through randomized search with cross-validation, a technique proven to be effective for complex models that have many hyperparameters [4].

We propose to evaluate the performance of our Hamiltonian-inspired method in comparison to XGBoost, an established benchmark in credit scoring [23]. Evaluating the robustness of both models in out-of-time scenarios is essential for effective credit risk assessment, as well as analyzing the trade-offs among model complexity, interpretability, and predictive accuracy in the field of credit scoring [13].

IV. RESULTS AND DISCUSSION

The first part of the experiment has been to find the best hyperparameters for both algorithms. In Table I we can see the best hyperparameters found for both algorithms.

TABLE I
BEST HYPERPARAMETERS FOUND FOR EXPERIMENTS

Neural Network with Hamiltonian approach	
Regulation coefficient, λ	0.01
Learning rate	0.01
Hidden dimensions	[128, 64]
Dropout rate	0.2
Activation function	Leaky ReLu
XGBoost algorithm	
Learning rate	0.3
Max depth	7
Min child weight	1
Number of estimators	300

Table II presents the comparative results of our Hamiltonian-inspired methodology and the XGBoost algorithm across three different Out-of-Time (OOT) datasets (FM12, FM36, and FM60). XGBoost consistently displays higher accuracy, precision, recall, and F1 scores across all temporal horizons. In the FM12 dataset, XGBoost achieves an accuracy of 0.9871, whereas the Hamiltonian approach scores 0.8047. This trend persists in FM36 and FM60, suggesting XGBoost's improved results in more common binary classification metrics.

The Hamiltonian-inspired method offers markedly superior Area Under the Curve (AUC) scores across all datasets. FM12 reaches an AUC of 0.8027, whereas XGBoost achieves 0.6072. This pattern is uniform in FM36 (0.764 compared to 0.6221) and FM60 (0.6974 compared to 0.6665). Higher AUC scores indicate that our approach has better discriminative capability and ranking competence, which is critical in credit scoring applications [18]. The AUC values calculated by the XGBoost algorithm are close to 0.5, suggesting that, despite its accuracy, its capacity for discrimination in unseen data is very close to randomness. Our approach implies more consistency in performance across different time horizons. The fluctuation in its metrics from FM12 to FM60 is minimal, with AUC scores varying from 0.8027 to 0.6974. XGBoost demonstrates superior initial performance; however, it indicates a significant decline in accuracy and recall over extended time periods. XGBoost has consistent high precision across all datasets (ranging from 0.9948 to 0.9796), yet demonstrates a gradual decline in recall over time (from 0.9827 to 0.9319). The Hamiltonian approach, despite exhibiting lower absolute values, reveals a more balanced precision-recall trade-off, which is important in the context of imbalanced datasets frequently found in credit scoring [32]. The consistent stability of the metrics in the Hamiltonian approach across different time windows suggests better generalization to future, unseen data. This part is particularly important for credit risk evaluation, as model performance typically declines over time due to concept drift [32].

Although XGBoost excels in standard metrics, the higher AUC scores of the Hamiltonian method suggest its potential advantage in ranking borrowers by risk. This is critical in practical credit scoring problems, where the rank-ordering of risk frequently holds greater significance than binary classification [35].

A. Statistical Significance and Ablation Analysis

To verify the robustness of our approach, we conducted statistical significance testing and ablation studies. Table III shows the statistical significance of AUC improvements using a paired t-test, demonstrating that our performance advantages are not due to random variation.

Additionally, we performed an ablation study to identify which components of our Hamiltonian-inspired approach contribute most to its performance. Table IV shows the impact of removing different components on the FM12 dataset.

This ablation study demonstrates that both the Hamiltonian normalization and momentum updates contribute significantly to the model's discriminative power, with their combined effect representing a substantial improvement over standard neural network approaches.

V. LIMITATIONS AND FUTURE WORK

While our Hamiltonian-inspired approach shows promising results, particularly in discriminative power and temporal stability, several important limitations must be acknowledged. The Hamiltonian optimization approach incurs additional computational costs compared to standard optimization techniques, potentially limiting its application in extremely high-dimensional or time-sensitive settings where resource constraints are significant. Furthermore, though our approach aims to capture underlying dynamics, the resulting models still face interpretability challenges inherent to neural networks—a particularly crucial consideration in the highly regulated financial sector where model transparency is often mandated by regulatory frameworks. The effectiveness of our approach also depends on appropriate settings for multiple hyperparameters, including regularization intensity and momentum coefficients, requiring careful tuning that demands domain expertise and computational resources for optimization. Additionally, while the Freddie Mac dataset provides comprehensive temporal coverage, our approach would benefit from testing across a wider variety of financial contexts and markets to establish broader applicability across different economic conditions and regulatory environments. Future work should address these limitations through several promising directions. Integration with traditional approaches could yield hybrid models that combine the discriminative power of our Hamiltonian-inspired methodology with the accuracy metrics of techniques like XGBoost, potentially offering a more balanced performance profile across different evaluation criteria. Interpretability enhancements through the development of visualization tools and explanation methods specific to Hamiltonian neural networks would make them more amenable to regulatory scrutiny.

TABLE II
COMPARATIVE RESULTS

OOT dataset	FM12 (12 months)	FM36 (36 months)	FM60 (60 months)
Hamiltonian inspired approach			
Mean accuracy	0.8047 $\pm$ 0.0021	0.7638 $\pm$ 0.0010	0.6975 $\pm$ 0.0003
Mean F1 Score	0.8012 $\pm$ 0.0000	0.7608 $\pm$ 0.0008	0.6968 $\pm$ 0.0003
Mean AUC score	0.8027 $\pm$ 0.0000	0.764 $\pm$ 0.0000	0.6974 $\pm$ 0.0000
Mean Precision	0.8201 $\pm$ 0.0000	0.777 $\pm$ 0.0000	0.6994 $\pm$ 0.0000
Mean Recall	0.8381 $\pm$ 0.0001	0.7638 $\pm$ 0.0000	0.6973 $\pm$ 0.0000
XGBoost algorithm			
Accuracy	0.9871	0.9663	0.9319
Precision	0.9948	0.9831	0.9796
Recall	0.9827	0.9663	0.9319
F1 Score	0.9882	0.9745	0.9542
AUC	0.6072	0.6221	0.6665

TABLE III
STATISTICAL SIGNIFICANCE OF AUC IMPROVEMENTS

Dataset	AUC Diff.	p-value	Significant?
FM12	0.1955	$p < 0.001$	Yes
FM36	0.1419	$p < 0.001$	Yes
FM60	0.0309	$p < 0.05$	Yes

TABLE IV
ABLATION STUDY RESULTS ON FM12 DATASET

Model Variant	AUC	% Decrease
Full Hamiltonian Model	0.8027	-
Without Hamiltonian Normalization	0.7645	4.76%
Without Momentum Updates	0.7412	7.66%
Standard Neural Network	0.7103	11.51%

and increase their practical utility in decision-making contexts where justification of outcomes is required. Developing automatic hyperparameter optimization specifically designed for Hamiltonian-inspired models could reduce the expertise required for effective implementation, democratizing access to these advanced techniques for a broader range of financial institutions. Extended validation across diverse financial datasets spanning different markets, economic cycles, and default rates would help establish the generalizability of our approach under varying conditions, strengthening confidence in its robustness. Finally, exploring other physics-inspired concepts such as energy landscapes and phase transitions could further enhance model stability during economic regime changes, potentially creating even more resilient models that maintain performance during periods of market stress or structural economic shifts. These future research directions highlight the potential for physics-inspired approaches to create a new paradigm in financial risk modeling that prioritizes stability and discrimination capabilities over conventional accuracy metrics, ultimately leading to more reliable risk assessment tools for the financial industry.

VI. CONCLUSIONS

This paper introduces a novel Hamiltonian-inspired neural network approach to credit scoring that demonstrates superior discriminative power and temporal stability compared to

traditional methods. Our approach leverages principles from physics to better capture the complex dynamics of credit risk evolution across time periods. The empirical evidence from the Freddie Mac Single-Family Loan-Level Dataset reveals a critical insight: while XGBoost achieves higher conventional accuracy metrics, our Hamiltonian-inspired methodology provides consistently better risk ranking capabilities—a crucial factor in practical credit scoring applications where discriminative power often holds greater importance than binary classification accuracy. Across all time horizons, we observed significantly higher AUC scores (0.8027 vs. 0.6072 in FM12, 0.764 vs. 0.6221 in FM36, and 0.6974 vs. 0.6665 in FM60), with statistical analysis confirming these improvements are not due to random variation. Furthermore, our approach demonstrates greater stability in performance metrics across different time windows, with minimal performance degradation compared to conventional techniques, and a balanced precision-recall trade-off that better handles the class imbalance inherent in credit scoring applications. These results suggest that incorporating physical system principles into financial risk modeling offers significant advantages for applications requiring long-term stability and accurate risk stratification. The practical implications extend beyond academic interest—financial institutions could potentially achieve more reliable risk assessments across changing economic conditions, leading to better-informed lending decisions and improved portfolio management. Beyond the immediate application to credit scoring, this research contributes to the broader interdisciplinary dialogue between physics, finance, and machine learning, demonstrating how principles from one field can provide novel perspectives and solutions to challenges in another, opening avenues for future cross-disciplinary research and application. As financial systems grow increasingly complex and interconnected, approaches that can maintain stability and accuracy across changing conditions become more valuable. Our Hamiltonian-inspired methodology represents a step toward more robust and reliable financial risk assessment, offering both theoretical insights and practical tools for an evolving financial landscape. Future work should focus on addressing the limitations identified, particularly around interpretability and computational complexity, while exploring hybrid approaches that combine

the strengths of our method with traditional techniques to further enhance practical applicability in regulated financial environments.

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